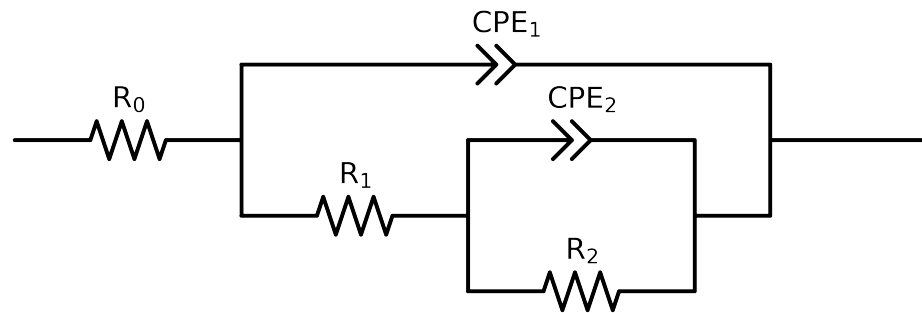


# Comparative EIS Kinetics Report

**Model Structure:** R0-p(R1-p(R2,CPE2),CPE1)  
**Used data:** altered data, first 1 points removed and points with  $freq > 1e + 05$  and  $freq < 1e - 03$  removed



| Element  | Unit                        | Bounds                   | 7_<br>Bare_<br>AA2024_<br>6 |
|----------|-----------------------------|--------------------------|-----------------------------|
| $R_0$    | $\Omega$                    | $[1.0e + 00, 1.0e + 03]$ | $1.14e + 02 \pm 3.1e - 01$  |
| $R_1$    | $\Omega$                    | $[1.0e + 00, 1.0e + 08]$ | $1.38e + 04 \pm 3.6e + 02$  |
| $R_2$    | $\Omega$                    | $[1.0e + 00, 1.0e + 12]$ | $3.60e + 04 \pm 2.1e + 03$  |
| $Q_2$    | $\Omega^{-1} \text{ sec}^a$ | $[1.0e - 09, 1.0e - 02]$ | $1.06e - 04 \pm 3.8e - 06$  |
| $n_2$    | -                           | $[5.0e - 01, 1.0e + 00]$ | $7.37e - 01 \pm 2.4e - 02$  |
| $Q_1$    | $\Omega^{-1} \text{ sec}^a$ | $[1.0e - 09, 1.0e - 03]$ | $2.66e - 05 \pm 2.8e - 07$  |
| $n_1$    | -                           | $[5.0e - 01, 1.0e + 00]$ | $8.14e - 01 \pm 1.9e - 03$  |
| $\chi^2$ | -                           | -                        | $1.499e - 04$               |

Analysis Results: 7\_Bare\_AA2024\_6

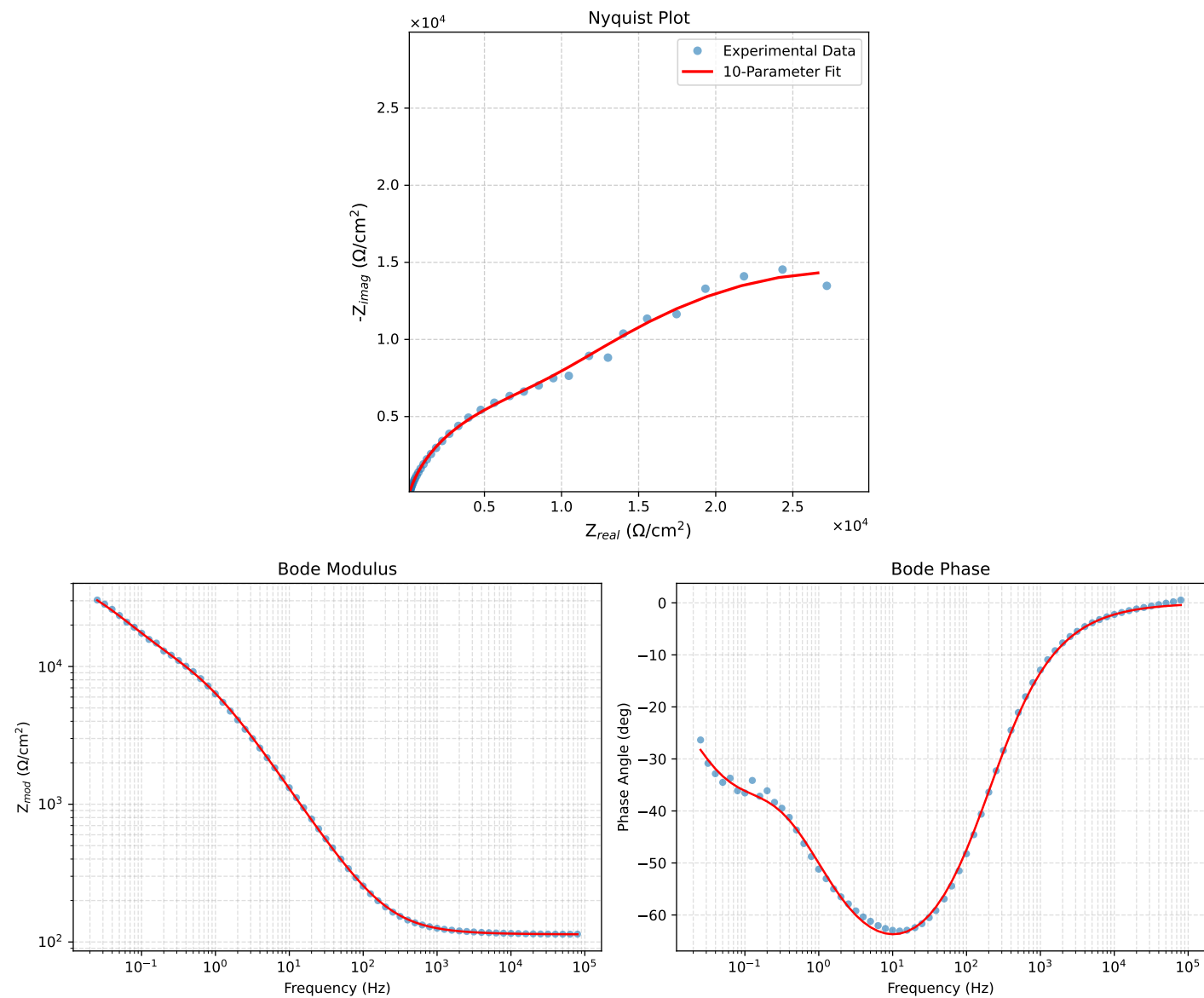


Figure 1: EIS Fit Results for 7\_Bare\_AA2024.6

# Fitted Data: 7\_Bare\_AA2024\_6

| Frequency (Hz) | $Z_{real} (\Omega)$ | $-Z_{imag} (\Omega)$ | $Z_{mod} (\Omega)$ | $Z_{phase} (^{\circ})$ |
|----------------|---------------------|----------------------|--------------------|------------------------|
| 7.9512e+04     | 1.1380e+02          | 8.3007e-01           | 1.1380e+02         | -0.4179                |
| 6.3105e+04     | 1.1385e+02          | 1.0018e+00           | 1.1386e+02         | -0.5041                |
| 5.0215e+04     | 1.1392e+02          | 1.2065e+00           | 1.1392e+02         | -0.6068                |
| 3.9902e+04     | 1.1399e+02          | 1.4547e+00           | 1.1400e+02         | -0.7311                |
| 3.1699e+04     | 1.1408e+02          | 1.7542e+00           | 1.1409e+02         | -0.8810                |
| 2.5137e+04     | 1.1419e+02          | 2.1186e+00           | 1.1421e+02         | -1.0629                |
| 1.9980e+04     | 1.1432e+02          | 2.5538e+00           | 1.1435e+02         | -1.2797                |
| 1.5879e+04     | 1.1448e+02          | 3.0787e+00           | 1.1452e+02         | -1.5405                |
| 1.2598e+04     | 1.1467e+02          | 3.7167e+00           | 1.1473e+02         | -1.8564                |
| 1.0020e+04     | 1.1490e+02          | 4.4778e+00           | 1.1499e+02         | -2.2317                |
| 7.9282e+03     | 1.1519e+02          | 5.4173e+00           | 1.1531e+02         | -2.6927                |
| 6.3079e+03     | 1.1552e+02          | 6.5247e+00           | 1.1570e+02         | -3.2327                |
| 5.0347e+03     | 1.1592e+02          | 7.8380e+00           | 1.1618e+02         | -3.8683                |
| 3.9931e+03     | 1.1641e+02          | 9.4644e+00           | 1.1679e+02         | -4.6481                |
| 3.1441e+03     | 1.1702e+02          | 1.1496e+01           | 1.1759e+02         | -5.6103                |
| 2.5195e+03     | 1.1771e+02          | 1.3764e+01           | 1.1851e+02         | -6.6693                |
| 2.0008e+03     | 1.1857e+02          | 1.6602e+01           | 1.1973e+02         | -7.9705                |
| 1.5807e+03     | 1.1964e+02          | 2.0109e+01           | 1.2132e+02         | -9.5410                |
| 1.2536e+03     | 1.2091e+02          | 2.4279e+01           | 1.2333e+02         | -11.3539               |
| 1.0007e+03     | 1.2240e+02          | 2.9159e+01           | 1.2583e+02         | -13.3994               |
| 7.9003e+02     | 1.2429e+02          | 3.5334e+01           | 1.2922e+02         | -15.8693               |
| 6.2881e+02     | 1.2651e+02          | 4.2532e+01           | 1.3346e+02         | -18.5828               |
| 5.0223e+02     | 1.2914e+02          | 5.1049e+01           | 1.3886e+02         | -21.5695               |
| 4.0015e+02     | 1.3234e+02          | 6.1388e+01           | 1.4589e+02         | -24.8846               |
| 3.1550e+02     | 1.3641e+02          | 7.4443e+01           | 1.5540e+02         | -28.6217               |
| 2.5202e+02     | 1.4109e+02          | 8.9317e+01           | 1.6698e+02         | -32.3362               |
| 2.0032e+02     | 1.4687e+02          | 1.0757e+02           | 1.8205e+02         | -36.2202               |
| 1.5801e+02     | 1.5416e+02          | 1.3035e+02           | 2.0189e+02         | -40.2157               |
| 1.2556e+02     | 1.6279e+02          | 1.5697e+02           | 2.2614e+02         | -43.9573               |
| 9.9734e+01     | 1.7333e+02          | 1.8904e+02           | 2.5647e+02         | -47.4817               |
| 7.9449e+01     | 1.8604e+02          | 2.2705e+02           | 2.9354e+02         | -50.6696               |
| 6.2919e+01     | 2.0203e+02          | 2.7390e+02           | 3.4035e+02         | -53.5880               |
| 4.9867e+01     | 2.2165e+02          | 3.3005e+02           | 3.9757e+02         | -56.1159               |
| 3.8422e+01     | 2.4920e+02          | 4.0651e+02           | 4.7681e+02         | -58.4909               |
| 3.1250e+01     | 2.7626e+02          | 4.7916e+02           | 5.5310e+02         | -60.0345               |
| 2.4934e+01     | 3.1250e+02          | 5.7300e+02           | 6.5268e+02         | -61.3929               |
| 2.0032e+01     | 3.5593e+02          | 6.8075e+02           | 7.6819e+02         | -62.3971               |
| 1.5625e+01     | 4.1780e+02          | 8.2651e+02           | 9.2610e+02         | -63.1833               |
| 1.2467e+01     | 4.8880e+02          | 9.8407e+02           | 1.0988e+03         | -63.5860               |
| 9.9734e+00     | 5.7657e+02          | 1.1668e+03           | 1.3014e+03         | -63.7028               |
| 7.9719e+00     | 6.8695e+02          | 1.3805e+03           | 1.5420e+03         | -63.5450               |
| 6.3516e+00     | 8.2764e+02          | 1.6318e+03           | 1.8297e+03         | -63.1064               |
| 5.0134e+00     | 1.0128e+03          | 1.9333e+03           | 2.1825e+03         | -62.3526               |
| 3.9860e+00     | 1.2389e+03          | 2.2661e+03           | 2.5826e+03         | -61.3333               |
| 3.1758e+00     | 1.5183e+03          | 2.6347e+03           | 3.0409e+03         | -60.0464               |
| 2.5161e+00     | 1.8724e+03          | 3.0496e+03           | 3.5785e+03         | -58.4515               |
| 1.9955e+00     | 2.3029e+03          | 3.4934e+03           | 4.1842e+03         | -56.6073               |
| 1.5879e+00     | 2.8106e+03          | 3.9513e+03           | 4.8490e+03         | -54.5751               |
| 1.2581e+00     | 3.4160e+03          | 4.4264e+03           | 5.5912e+03         | -52.3410               |
| 9.9777e-01     | 4.1036e+03          | 4.8954e+03           | 6.3879e+03         | -50.0285               |
| 7.9274e-01     | 4.8603e+03          | 5.3480e+03           | 7.2266e+03         | -47.7356               |
| 6.2903e-01     | 5.6824e+03          | 5.7879e+03           | 8.1110e+03         | -45.5267               |
| 4.9888e-01     | 6.5536e+03          | 6.2191e+03           | 9.0348e+03         | -43.5000               |
| 3.9860e-01     | 7.4326e+03          | 6.6411e+03           | 9.9673e+03         | -41.7812               |
| 3.1672e-01     | 8.3670e+03          | 7.0970e+03           | 1.0971e+04         | -40.3052               |
| 2.5148e-01     | 9.3460e+03          | 7.5995e+03           | 1.2046e+04         | -39.1153               |
| 1.9998e-01     | 1.0376e+04          | 8.1615e+03           | 1.3202e+04         | -38.1866               |
| 1.5879e-01     | 1.1497e+04          | 8.8027e+03           | 1.4480e+04         | -37.4399               |
| 1.2601e-01     | 1.2736e+04          | 9.5244e+03           | 1.5903e+04         | -36.7912               |
| 1.0016e-01     | 1.4114e+04          | 1.0310e+04           | 1.7478e+04         | -36.1475               |
| 7.9503e-02     | 1.5682e+04          | 1.1146e+04           | 1.9239e+04         | -35.4038               |
| 6.3140e-02     | 1.7455e+04          | 1.1989e+04           | 2.1176e+04         | -34.4832               |
| 5.0123e-02     | 1.9456e+04          | 1.2791e+04           | 2.3284e+04         | -33.3222               |
| 3.9833e-02     | 2.1667e+04          | 1.3485e+04           | 2.5521e+04         | -31.8982               |
| 3.1638e-02     | 2.4076e+04          | 1.4014e+04           | 2.7858e+04         | -30.2014               |
| 2.5126e-02     | 2.6631e+04          | 1.4316e+04           | 3.0235e+04         | -28.2610               |